ChargeGuard AI - Day 2 Report

PayPal AI Hackathon 2026 | Report date: 4 Nov 2026 | Deadline: 12 Nov 2026, 12:00 PT

1. Scope of Day 2

The build plan treats **Days 1-2** as one phase: *credentials, Transaction Search permission, webhook receiver + signature verification, public tunnel, seeder, schema.* Most of it was finished in the first session, so this report covers what is done, what Day 2 still has to close out, and the order to do it in.

2. Day 1-2 checklist

Plan item	Status	Detail
Sandbox credentials	**DONE**	New US Business + US Personal accounts and a US REST app. The first India-based accounts failed at buyer checkout, so they were replaced.
Webhook receiver + signature verification	**DONE**	POST /webhooks/paypal calls PayPal's verify-webhook-signature before storing anything. Invalid or unsigned requests return 400.
Idempotent event storage	**DONE**	webhook_events.paypal_event_id is the key, so PayPal retries are ignored. Covered by the one runnable test.
Public tunnel	**DONE (temporary)**	cloudflared quick tunnel. The URL changes on every restart; Render replaces it for the final demo.
Schema (7 tables)	**DONE**	Status columns are constrained in SQL (case states, action states and types).
Seeder - subscription	**DONE**	npm run seed creates product, plan ($9.99/month) and subscription with custom_id. One manual buyer approval is needed.
End-to-end proof	**DONE**	After approval, PayPal returned ACTIVE and 8 signed webhooks were stored, including SUBSCRIPTION.ACTIVATED and SALE.COMPLETED.
Transaction Search permission	**TO CONFIRM**	Must be enabled on the NEW app. It can take up to 9 hours to take effect, so enable it first thing.
Seeder - buyer-side disputes	**NOT STARTED**	Needs buyer credentials and a PayPal-Auth-Assertion header. The plan budgets half a day. Required for the dispute demo path.
AI provider key	**NOT SET**	AI_API_KEY is empty. Not needed until the AI jobs (day 5) but should be ready.

3. Day 2 task list (in order)

#	Task	Why / how to know it is done
1	Enable Transaction Search on the new US app	Starts the up-to-9-hour clock. Done when the dashboard shows the permission on.
2	Repo cleanup on GitHub	Delete the two accidental workflow commits (webpack.yml, Deno). Decide how to handle plan.md, which is gone from main but still in commit 19cfcc7.

#	Task	Why / how to know it is done
3	Seeder: buyer-side dispute creation	Create the buyer API credentials, add the PayPal-Auth-Assertion header, and open a sandbox dispute against a seeded sale. Done when a CUSTOMER.DISPUTE.CREATED webhook is stored.
4	Seeder: sandbox dispute helpers	Change reason, update status, settle in merchant favour. These drive the demo's winning ending.
5	Add the Gemini key	Fill AI_API_KEY in .env; one test call to generateJson() proves the AI client works.
6	Rotate the old India app secret	Housekeeping - those keys were exposed in a screenshot and in chat.
7	Commit and push	Check git status first so .env, screenshots and PDFs are not added by accident.

4. Risks to watch

Risk	Mitigation
Tunnel URL changes when cloudflared restarts	Keep the terminal open; if the URL changes, edit the webhook URL in the dashboard. Move to Render before recording the video.
Webhooks arrive out of order (ACTIVATED can land after the first SALE.COMPLETED)	Day 3 dispatcher upserts on subscription ID and re-fetches state from the API.
Transaction Search is not real-time	Treat it as backfill only; webhooks are the source of truth.
Stale env in a long-running dev server	The server reads .env only at startup. Restart it after any .env change.
Public repo shows history	plan.md and early commits remain visible in history unless it is rewritten.

5. Commands

```
cd server && npm run dev      # API on :3000 (restart after .env edits)
cloudflared tunnel --url http://localhost:3000
npm run seed                  # new sandbox subscription + approval link
npm test                      # idempotency check
```

After Day 2: Days 3-4 build the case engine - a webhook dispatcher into the subscriptions and transactions tables, the SQL rules layer (duplicate charge, failed payment), dispute ingestion and the deadline clock.